Doping Stability Report -- Zn in NaCoO2 at 300 C

Generated: 2026-05-30 09:12   Host: NaCoO2   Dopant: Zn

Executive Summary

Zn preferentially occupies the Co layer. The formation energy difference between the two layers is +2.930 eV, strongly favouring the Co site. MD confirms this preference: on the Co site the dopant MSD plateau slope is 0.00173 A^2/ps (stable), compared to -0.00027 A^2/ps on the Na site. The Na-site E_f is approximate due to a charge surplus of +1 that cannot be modelled in CHGNet.

Site	E_f (eV)	Charge	Compensation	MSD pass	Coord pass	Verdict
Co	+0.055	-1	No	PASS	PASS	STABLE
Na	+2.985	+1	Yes	PASS	FAIL	STRUCTURAL COLLAPSE

Zn at the Co layer -- STABLE

Charge situation

Zn replaces Co with a charge mismatch of -1. To restore charge neutrality, 1 Na atom(s) were removed from the supercell (furthest from the dopant site). The formation energy accounts for the Na-vacancy term.

Formation Energy (thermodynamic stability)

Formation energy E_f	**+0.0553 eV**	PASS	*< 1.0 eV to pass*

E_f = +0.055 eV. This value is small and positive, indicating that Zn incorporation is marginally feasible under equilibrium conditions at the Co site.

Molecular Dynamics Stability (300 C, 25 ps)

MSD slope (late-time)	**0.00173 A^2/ps**	PASS	*< 0.005 to pass (plateau = stable)*
Final MSD	**0.207 A^2**	----	
Max displacement	**1.060 A**	----	
Coordination (min/mean/max)	**6 / 6.0 / 6**	PASS	*relaxed = 6, tolerance +/-1*
Mean bond length (MD vs relaxed)	**2.054 A (relaxed: 2.0**	----	
Lattice volume change	**+0.00 %**	PASS	*< +/-5% to pass*

Late-time MSD slope = 0.00173 A^2/ps. The slope is small and below the migration threshold -- the dopant is site-stable. Final MSD = 0.207 A^2; maximum displacement from starting position = 1.06 A.

Coordination was exactly 6 throughout the entire trajectory -- a near-perfect result indicating the dopant held its local environment rigidly. Mean nearest-neighbour distance during MD = 2.054 A.

Volume change = +0.00%. The host lattice is mechanically intact.

VERDICT: STABLE

Zn is thermodynamically favorable and kinetically stable on the Co site at 300 C. All four stability criteria pass: formation energy is favorable, the dopant does not migrate during MD, its coordination environment is preserved, and the host lattice volume is unchanged.

? WARNING: Charge deficit (-1) cannot be compensated in CHGNet (would require adding Na interstitials or oxidising Co). Running single substitution without compensation. E_f is approximate.

Zn at the Na layer -- STRUCTURAL COLLAPSE

Charge situation

Zn replaces Na with a charge surplus of +1. CHGNet cannot model this compensation (would require Co oxidation or electron addition). The formation energy below is approximate.

Formation Energy (thermodynamic stability)

Formation energy E_f	**+2.9853 eV**	FAIL	*< 1.0 eV to pass*

E_f = +2.985 eV (approximate -- charge surplus not modelled in CHGNet). This value is large and positive, indicating that Zn incorporation is thermodynamically unfavorable under equilibrium conditions at the Na site.

Molecular Dynamics Stability (300 C, 25 ps)

MSD slope (late-time)	**-0.00027 A^2/ps**	PASS	*< 0.005 to pass (plateau = stable)*
Final MSD	**0.982 A^2**	----	
Max displacement	**1.553 A**	----	
Coordination (min/mean/max)	**3 / 5.6 / 7**	FAIL	*relaxed = 6, tolerance +/-1*
Mean bond length (MD vs relaxed)	**2.164 A (relaxed: 2.1**	----	
Lattice volume change	**+0.00 %**	PASS	*< +/-5% to pass*

Late-time MSD slope = -0.00027 A^2/ps. The negative slope is a flat plateau -- essentially no migration detected. Final MSD = 0.982 A^2; maximum displacement from starting position = 1.55 A.

Coordination fluctuated between 3 and 7 (mean 5.6), deviating by more than +/-1 from the relaxed value (6). This may indicate structural distortion, or that the bond-length cutoff is too close to the actual bond length -- check the coordination_cutoff_A setting. Mean nearest-neighbour distance during MD = 2.164 A.

Volume change = +0.00%. The host lattice is mechanically intact.

VERDICT: STRUCTURAL COLLAPSE

The coordination criterion failed while MSD and volume both pass, suggesting a possible cutoff artifact rather than a true structural collapse. The Zn atom barely moves (MSD = 0.982 A^2) but the coordination count fluctuates because the bond length at the Na site is close to the analysis cutoff. Consider increasing coordination_cutoff_A and re-running analysis.

Site Preference Comparison

The Co site is preferred by 2.930 eV over the Na site. Formation energy: +0.055 eV (Co) vs +2.985 eV (Na).

Zn preferentially occupies the Co layer. The formation energy difference between the two layers is +2.930 eV, strongly favouring the Co site. MD confirms this preference: on the Co site the dopant MSD plateau slope is 0.00173 A^2/ps (stable), compared to -0.00027 A^2/ps on the Na site. The Na-site E_f is approximate due to a charge surplus of +1 that cannot be modelled in CHGNet.

Caveats and Limitations

? **CHGNet is charge-neutral.**

The potential does not model oxidation states, polaron formation, or electron/hole localisation. Formation energies for aliovalent dopants (charge mismatch != 0) are approximate even when structural compensation defects are included.

? **No Freysoldt/Kumagai correction.**

Electrostatic finite-size corrections for charged defects require the host dielectric tensor and the DFT electrostatic potential, neither of which is available from a neural-network potential. For publication-quality E_f of charged defects, follow up with DFT using doped or pymatgen-analysis-defects.

? **Metal-rich reference state.**

Chemical potentials are computed from elemental ground-state structures. Synthesis under oxide-rich or other conditions shifts all E_f values by additive constants; the relative ordering of sites is unaffected.

? **MD timescales are short (25 ps default).**

This is sufficient to assess local site stability and rapid migration, but cannot resolve slow diffusion mechanisms. A non-plateauing MSD confirms instability; a plateau does not guarantee stability on longer timescales.

? **Energy rankings are more trustworthy than absolute values.**

CHGNet is a universal MLIP trained on r2SCAN DFT. Formation-energy errors for transition-metal oxides are typically 50-150 meV/atom. Use relative orderings (which site is preferred?) as the primary result.